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Atezolizumab (Tecentriq) and Atezolizumab and Hyaluronidase-tqjs (Tecentriq Hybreza)

Policy History

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Review History

Definitions

Additional Information

Clinical Policy Bulletin Notes

Number: 0909

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Policy

Scope of Policy

This Clinical Policy Bulletin addresses atezolizumab (Tecentriq) and atezolizumab and hyaluronidase-tqjs (Tecentriq Hybreza) for commercial medical plans. For Medicare criteria, see Medicare Part B Criteria.

Note: Requires Precertification:

Precertification of atezolizumab (Tecentriq) and atezolizumab and hyaluronidase-tqjs (Tecentriq Hybreza) is required of all Aetna participating providers and members in applicable plan designs. For precertification of atezolizumab (Tecentriq), call (866) 752-7021 or fax (888) 267-3277. For Statement of Medical Necessity (SMN) precertification forms, see Specialty Pharmacy Precertification.

Note: Site of Care Utilization Management Policy applies for atezolizumab (Tecentriq) and atezolizumab and hyaluronidase-tqjs (Tecentriq Hybreza). For information on site of service, see Utilization Management Policy on Site of Care for Specialty Drug Infusions.

Atezolizumab (Tecentriq)

I. *Exclusions*

Aetna will not provide coverage for members who have experienced disease progression while on PD-1 (e.g., nivolumab [Opdivo], pembrolizumab [Keytruda], and cemiplimab [Libtayo]) or PD-L1 inhibitor (e.g., atezolizumab [Tecentriq], atezolizumab and hyaluronidase-tqjs (Tecentriq Hybreza), avelumab [Bavencio], durvalumab [Imfinzi]) therapy.

II. *Criteria for Initial Approval*

Aetna considers atezolizumab (Tecentriq) medically necessary for the following indications:

A. Non-small Cell Lung Cancer (NSCLC)

1. For treatment of recurrent, advanced or metastatic non-small cell lung cancer when there are no EGFR exon 19 deletions or L858R mutations or ALK rearrangements (unless testing is not feasible due to insufficient tissue) and *any* of the following criteria are met:

   a. The requested medication will be used as continued maintenance therapy as a single agent

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or in combination with bevacizumab; *or*

b. The requested medication will be used as first line or subsequent therapy in combination with chemotherapy with or without bevacizumab; *or*
c. The requested medication will be used as first line therapy for PD-L1 expression positive (greater than or equal to 50%) tumors as a single agent;

2. For treatment of stage II to III non-small cell lung cancer that is PD-L1 positive as single agent adjuvant therapy;
3. For treatment of recurrent, advanced or metastatic non-small cell lung cancer as single agent subsequent therapy;

B. Small Cell Lung Cancer (SCLC)

For treatment of small cell lung cancer when the requested medication will be used as initial treatment in combination with etoposide and carboplatin (followed by single agent maintenance) for extensive-stage disease;

C. Hepatocellular Carcinoma (HCC)

1. For treatment of unresectable, inoperable, metastatic, or disease with extensive liver tumor burden HCC when the requested medication will be used as initial treatment in combination with bevacizumab;
2. In combination with bevacizumab for adjuvant treatment following resection or ablation;

D. Melanoma

For the treatment of BRAF V600 mutation-positive unresectable or metastatic melanoma when the requested medication will be used in combination with cobimetinib (Cotellic) and vemurafenib (Zelboraf);

E. Mesothelioma

For the subsequent treatment of peritoneal mesothelioma, pericardial mesothelioma, or tunica vaginalis testis mesothelioma when used in combination with bevacizumab;

F. Alveolar Soft Part Sarcoma (ASPS)

For the treatment of unresectable or metastatic alveolar soft part sarcoma when used as a single agent;

G. Cervical Cancer

For the treatment of persistent, recurrent or metastatic small cell neuroendocrine carcinoma of the cervix (NECC) when used in combination with etoposide and either cisplatin or carboplatin

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cervix (NECC) when used in combination with etoposide and either cisplatin or carboplatin.

Aetna considers all other indications as experimental, investigational, or unproven.

III. *Continuation of Therapy*

A. Adjuvant treatment of Hepatocellular Carcinoma (HCC) and Non-Small Cell Lung Cancer (NSCLC)

Aetna considers continuation of atezolizumab (Tecentriq) therapy (up to 12 months total) medically necessary in members requesting reauthorization of adjuvant treatment of hepatocelluar carcinoma and non-small cell lung cancer who have not experienced disease recurrence or an unacceptable toxicity.

B. All other indications

Aetna considers continuation of atezolizumab (Tecentriq) therapy in members requesting reauthorization for an indication listed in the criteria for initial approval section when there is no evidence of unacceptable toxicity or disease progression while on the current regimen.

Atezolizumab and Hyaluronidase-tqjs (Tecentriq Hybreza)

I. *Exclusions*

Aetna will not provide coverage for members who have experienced disease progression while on PD-1 (e.g., nivolumab [Opdivo], pembrolizumab [Keytruda], and cemiplimab [Libtayo]) or PD-L1 inhibitor (e.g., atezolizumab [Tecentriq], atezolizumab and hyaluronidase-tqjs (Tecentriq Hybreza), avelumab [Bavencio], durvalumab [Imfinzi]) therapy.

II. *Criteria for Initial Approval*

Aetna considers atezolizumab and hyaluronidase-tqjs (Tecentriq Hybreza) medically necessary for the following indications:

A. Non-small Cell Lung Cancer (NSCLC)

1. For treatment of recurrent, advanced or metastatic non-small cell lung cancer when there are no EGFR exon 19 deletions or L858R mutations or ALK rearrangements (unless testing is not feasible due to insufficient tissue) and *any* of the following criteria are met:

   a. The requested medication will be used as continued maintenance therapy as a single agent or in combination with bevacizumab; *or*
   b. The requested medication will be used as first line or subsequent therapy in combination with chemotherapy with or without bevacizumab; *or*

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c. The requested medication will be used as first line therapy for PD-L1 expression positive (greater than or equal to 50%) tumors as a single agent;

2. For treatment of stage II to III non-small cell lung cancer that is PD-L1 positive as single agent adjuvant therapy;
3. For treatment of recurrent, advanced or metastatic non-small cell lung cancer as single agent subsequent therapy;

B. Small Cell Lung Cancer (SCLC)

For treatment of SCLC when the requested medication will be used as initial treatment in combination with etoposide and carboplatin for extensive-stage disease;

C. Hepatocellular Carcinoma (HCC)

1. In combination with bevacizumab for first-line treatment of unresectable or metastatic HCC; *or*
2. In combination with bevacizumab for adjuvant treatment following resection or ablation;

D. Melanoma

For the treatment of BRAF V600 mutation-positive unresectable or metastatic melanoma when the requested medication will be used in combination with cobimetinib (Cotellic) and vemurafenib (Zelboraf);

E. Mesothelioma

For the subsequent treatment of peritoneal mesothelioma, pericardial mesothelioma, or tunica vaginalis testis mesothelioma when used in combination with bevacizumab;

F. Alveolar Soft Part Sarcoma (ASPS)

For the treatment of unresectable or metastatic ASPS when used as a single agent;

G. Cervical Cancer

For the treatment of persistent, recurrent or metastatic small cell neuroendocrine carcinoma of the cervix (NECC) when used in combination with etoposide and either cisplatin or carboplatin (followed by single agent maintenance).

Aetna considers all other indications as experimental, investigational, or unproven.

III. *Continuation of Therapy*

A. Adjuvant treatment of Hepatocellular Carcinoma (HCC) or Non-Small Cell Lung Cancer (NSCLC)

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Aetna considers continuation of atezolizumab and hyaluronidase-tqjs (Tecentriq Hybreza) therapy (up to 12 months total) medically necessary in members requesting reauthorization of adjuvant

therapy of hepatocellular carcinoma or non-small cell lung cancer who have not experienced disease who have not experienced disease recurrence or an unacceptable toxicity.

B. All other indications

Aetna considers continuation of atezolizumab and hyaluronidase-tqjs (Tecentriq Hybreza) therapy in members requesting reauthorization for an indication listed in the criteria for initial approval section when there is no evidence of unacceptable toxicity or disease progression while on the current regimen.

Dosage and Administration

Atezolizumab (Tecentriq)

Atezolizumab (Tecentriq) is available as 840 mg/14 mL (60 mg/mL) and 1200 mg/20mL (60 mg/mL) solution in a single-dose vial to be administered as an intravenous (IV) infusion over 60 minutes until disease progression or unacceptable toxicity unless otherwise specified. If the first infusion is tolerated, subsequent infusions may be delivered over 30 minutes.

Non-Small Cell Lung Cancer (NSCLC)

- Adjuvant Treatment of NSCLC: Administer Tecentriq following resection and up to 4 cycles of platinum-based chemotherapy as 840 mg every 2 weeks. 1200 mg every 3 weeks or 1680 mg every 4 weeks for up to 1 year, unless there is disease recurrence or unacceptable toxicity.
- Metastatic NSCLC: Administer Tecentriq 840 mg every 2 weeks, 1200 mg every 3 weeks, or 1680 mg every 4 weeks.
- When administering with chemotherapy with or without bevacizumab, administer Tecentriq prior to chemotherapy and bevacizumab when given on the same day.

Hepatocellular Carcinoma (HCC)

Administer Tecentriq is 840 mg every 2 weeks, or 1200 mg every 3 weeks, or 1680 mg every 4 weeks. Administer Tecentriq prior to bevacizumab when given on the same day. Bevacizumab is administered at 15 mg/kg every 3 weeks.

Melanoma

Following completion of a 28 day cycle of cobimetinib and vemurafenib, administer Tecentriq 840 mg every 2 weeks, 1200 mg every 3 weeks, or 1680 mg every 4 weeks with cobimetinib 60 mg orally once

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every 2 weeks, 1200 mg every 3 weeks, or 1680 mg every 4 weeks with cobimetinib 60 mg orally once daily (21 days on and 7 days off) and vemurafenib 720 mg orally twice daily.

Small Cell Lung Cancer (SCLC)

Administer Tecentriq is 840 mg every 2 weeks, 1200 mg every 3 weeks, or 1680 mg every 4 weeks. When administering with carboplatin and etoposide, administer Tecentriq prior to chemotherapy when given on the same day.

Alveolar Soft Part Sarcoma (ASPS)

- Adults: Administer Tecentriq 840mg every 2 weeks, 1200 mg every 3 weeks, or 1680 mg every 4 weeks until disease progression or unacceptable toxicity.
- Pediatric individuals 2 years of age and older: 15 mg/kg (up to a maximum of 1200 mg) every 3 weeks until disease progression or unacceptable toxicity.

Refer to full prescribing information for Tecentriq for preparation and administration instructions and dosage modifications.

Source: Genentech, 2024a

Atezolizumab and Hyaluronidase-tqjs (Tecentriq Hybreza)

Atezolizumab and hyaluronidase-tqjs is supplied as Tecentriq Hybreza 1,875 mg atezolizumab and 30,000 units hyaluronidase per 15 mL (125 mg/2,000 units per mL) solution in a single-dose vial for subcutaneous injection.

- Tecentriq Hybreza has different recommended dosage and administration than intravenous atezolizumab products.
- Tecentriq Hybreza is not indicated for use in pediatric patients.
- Tecentriq Hybreza is for subcutaneous use in the thigh only.
- Do not administer Tecentriq Hybreza intravenously.
- The recommend dosage for Tecentriq Hybreza is 15 mL (1,875 mg atezolizumab and 30,000 units hyaluronidase) subcutaneously into the thigh over approximately 7 minutes every 3 weeks.
- Tecentriq Hybreza must be administered by a healthcare professional.

Non-small Cell Lung Cancer (NSCLC) Dosage

- Adjuvant setting: Administer Tecentriq Hybreza following resection up to 4 cycles of platinum-based chemotherapy every 3 weeks for up to 1 year.
- Metastatic setting: Administer Tecentriq Hybreza every 3 weeks.
- When administering with chemotherapy with or without bevacizumab, administer Tecentriq

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Hybreza prior to chemotherapy and bevacizumab when given on the same day.

Small-cell Lung Cancer (SCLC) Dosage

- Administer Tecentriq Hybreza every 3 weeks. When administering with carboplatin and etoposide, administer Tecentriq Hybreza prior to chemotherapy when given on the same day.

Hepatocellular Carcinoma (HCC) Dosage

- Administer Tecentriq Hybreza every 3 weeks.
- Administer Tecentriq Hybreza prior to bevacizumab when given on the same day. Bevacizumab is administered intravenously at 15 mg/kg every 3 weeks.

Melanoma Dosage

- Following completion of a 28-day cycle of cobimetinib and vemurafenib, administer Tecentriq Hybreza every 3 weeks with cobimetinib 60 mg orally once daily (21 days on /7days off) and vemurafenib 720 mg orally twice daily.

Alveolar Soft Part Sarcoma (ASPS) Dosage

- Administer Tecentriq Hybreza every 3 weeks.

Source: Genentech, 2024b

Experimental, Investigational, or Unproven

I. Aetna considers atezolizumab or atezolizumab and hyaluronidase-tqjs experimental, investigational, or unproven for all other indications including the following (not an all-inclusive list) due to insufficient evidence in the peer-reviewed literature:

- Acute myeloid leukemia
- Appendiceal adenocarcinoma
- Asymptomatic myeloma
- Bone plasmacytoma
- Cholangiocarcinoma
- Chronic lymphocytic leukemia
- Chronic myelomonocytic leukemia
- Fallopian tube carcinoma
- Gastrointestinal cancers (e.g., colorectal, esophageal, gastric, and gastro-esophageal junction

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cancers)
- Glioblastoma

- Hodgkin lymphoma
- Large cell neuroendocrine carcinoma
- Melanoma with central nervous system metastases
- Mesothelioma
- Metastatic neuroendocrine tumor of the intestines
- Multiple myeloma
- Myelodysplastic syndrome
- Myxoid/round cell liposarcoma
- Non-Hodgkin's lymphoma
- Ovarian carcinoma
- Pancreatic adenocarcinoma
- Paraganglioma
- Peritoneal carcinoma
- Pheochromocytoma
- Polycythemia vera
- Primary myelofibrosis
- Prostate cancer
- Renal cell carcinoma
- Small lymphocytic lymphoma
- Smoldering multiple myeloma
- Synovial sarcoma
- Thyroid carcinoma
- Urinary tract squamous cell carcinoma
- Uterine cancer
- Uveal melanoma.

II. Aetna considers atezolizumab or atezolizumab and hyaluronidase-tqjs with or without radiotherapy experimental, investigational, or unproven for the treatment of squamous cell carcinoma of the penis.

III. Aetna considers atezolizumab or atezolizumab and hyaluronidase-tqjs plus cobimetinib experimental, investigational, or unproven for the treatment of melanoma with central nervous system metastases.

IV. Aetna considers combined tiragolumab and atezolizumab or atezolizumab and hyaluronidase-tqjs experimental, investigational, or unproven for the treatment of solid tumors.

CPT Codes / HCPCS Codes / ICD-10 Codes

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CPT Codes / HCPCS Codes / ICD-10 Codes

Other CPT codes related to the CPB :

Code	Code Description
81235	EGFR (epidermal growth factor receptor) (eg, non-small cell lung cancer) gene analysis, common variants (eg, exon 19 LREA deletion, L858R, T790M, G719A, G719S, L861Q)
96413 - 96415	Chemotherapy administration, intravenous infusion technique
HCPCS codes covered if selection criteria are met:	
J9022	Injection, atezolizumab, 10 mg
Other HCPCS codes related to the CPB:	
***Cobimetinib (Cotellic), vemurafenib (Zelboraf), Tiragolumab* - no specific code**	
J8560	Etoposide; oral, 50 mg
J8565	Gefitinib, oral, 250 mg
J9023	Injection, avelumab, 10 mg
J9035	Injection, bevacizumab, 10 mg
J9045	Injection, carboplatin, 50 mg
J9060	Injection, cisplatin, powder or solution, 10 mg
J9119	Injection, cemiplimab-rwlc, 1 mg
J9173	Injection, durvalumab, 10 mg
J9181	Injection, etoposide, 10 mg
J9263	Injection, oxaliplatin, 0.5 mg
J9258	Injection, paclitaxel protein-bound particles (teva) not therapeutically equivalent to j9264, 1 mg
J9259	Injection, paclitaxel protein-bound particles (american regent) not therapeutically equivalent to j9264, 1 mg
J9264	Injection, paclitaxel protein-hyphenbound particles, 1 mg
J9267	Injection, paclitaxel, 1 mg
J9271	Injection, pembrolizumab, 1 mg
J9299	Injection, nivolumab, 1 mg
Q5107	Injection, bevacizumab-awwb, biosimilar, (mvasi), 10 mg
ICD-10 codes covered if selection criteria are met:	
C22.0	Liver cell carcinoma [unresectable or metastatic]
C34.00 - C34.92	Malignant neoplasm of bronchus and lung
C43.0 - C43.9	Malignant melanoma of skin [not covered for melanoma with central nervous system metastases]
C45.0 - C45.9	Mesothelioma

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C49.0 - C49.9	Malignant neoplasm of other connective and soft tissue [alveolar soft part sarcoma] [not
Code	**Code Description**
	covered for Myxoid/round cell liposarcoma and Synovial sarcoma]
C50.011 - C50.929	Malignant neoplasm of breast
C53.0 – C53.9	Malignant neoplasm of cervix uteri
ICD-10 codes not covered for indications listed in the CPB (not all inclusive):	
C11.00 - C80.2, D00.00 - D09.22	Solid cell tumors [not covered for combination with tiragolumab and atezolizumab]
C15.3 - C20	Malignant neoplasm of esophagus, stomach, small intestine (including duodenum), colon, rectosigmoid junction and rectum
C22.1 - C22.9	Malignant neoplasm of primary and intrahepatic bile ducts [cholangiocarcinoma]
C25.0 - C25.9	Malignant neoplasm of pancreas [pancreatic adenocarcinoma]
C44.529	Squamous cell carcinoma of skin of other part of trunk [penis]
C48.0 - C48.8	Malignant neoplasm of retroperitoneum and peritoneum
C4A.0 - C4A.9	Merkel cell carcinoma
C54.0 - C55	Malignant neoplasm of uterus
C56.1 - C56.9	Malignant neoplasm of ovary
C57.00 - C57.02	Malignant neoplasm of fallopian tube
C64.1 - C68.9	Malignant neoplasm of urinary tract
C69.30 - C69.32	Malignant neoplasm of choroid [uveal melanoma]
C69.40 - C69.42	Malignant neoplasm of ciliary body [uveal melanoma]
C71.0 - C71.9	Malignant neoplasm of brain [glioblastoma]
C72.0 - C72.9	Malignant neoplasm of central nervous system [glioblastoma]
C73	Malignant neoplasm of thyroid gland
C75.5	Malignant neoplasm of aortic body and other paraganglia [Paraganglioma]
C74.10 – C74.12	Malignant neoplasm of medulla of adrenal gland [pheochromocytoma]
C79.31	Secondary malignant neoplasm of brain [melanoma with central nervous system metastases]
C79.32	Secondary malignant neoplasm of cerebral meninges [melanoma with central nervous system metastases]
C79.40	Secondary malignant neoplasm of unspecified part of nervous system [melanoma with central nervous system metastases]
C79.49	Secondary malignant neoplasm of other parts of nervous system [melanoma with central nervous system metastases]

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C7A.00 - C7A.8	Malignant neuroendocrine tumors

Code	Code Description
C7B.8	Other secondary neuroendocrine tumors [metastatic neuroendocrine tumor of intestines]
C81.00 - C81.99	Hodgkin lymphoma
C82.00 - C91.92	Malignant neoplasm of lymphoid, hematopoietic and related tissue [non-Hodgkin's lymphoma]
C90.00 - C90.02	Multiple myeloma [asymptomatic myeloma, smoldering multiple myeloma]
C90.20 - C90.32	Extramedullary/solitary plasmacytoma
C91.10 - C91.12, C91.90 - C91.91	Chronic lymphoid leukemia [chronic lymphocytic leukemia/small lymphocytic lymphoma (CLL/SLL)]
C92.00 - C92.02	Acute myeloid leukemia
C93.10 - C93.12	Chronic myelomonocytic leukemia
D44.7	Neoplasm of uncertain behavior of aortic body and other paraganglia [Paraganglioma]
D45	Polycythemia vera
D46.0 - D46.9	Myelodysplastic syndrome
D75.81	Myelofibrosis [primary]

Background

U.S. Food and Drug Administration (FDA)-Approved Indications for Atezolizumab (Tecentriq)

- *Non-small cell lung cancer (NSCLC)*
    - Tecentriq, as a single-agent, is indicated as adjuvant treatment following resection and platinum-based chemotherapy for adult patients with stage II to IIIA non-small cell lung cancer (NSCLC) whose tumors have PD-L1 expression on ≥ 1% of tumor cells, as determined by an FDA-approved test;
    - Tecentriq, as a single-agent, is indicated for the first line treatment of adult patients with metastatic NSCLC whose tumors have high PD-L1 expression (PD-L1 stained ≥ 50% of tumor cells [TC ≥ 50%] or PD-L1 stained tumor-infiltrating immune cells [IC] covering ≥ 10% of the tumor area [IC ≥ 10%]), as determined by an FDA-approved test, with no EGFR or ALK genomic tumor aberrations;
    - Tecentriq, in combination with bevacizumab, paclitaxel, and carboplatin, is indicated for the

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- Tecentriq, in combination with bevacizumab, paclitaxel, and carboplatin, is indicated for the first-line treatment, of adult patients with metastatic non-squamous NSCLC with no EGFR or ALK genomic tumor aberrations;
- Tecentriq, in combination with paclitaxel protein-bound and carboplatin, is indicated for the first-line treatment of adult patients with metastatic non-squamous NSCLC with no EGFR or ALK genomic tumor aberrations;
- Tecentriq, as a single agent, is indicated for the treatment of adult patients with metastatic NSCLC who have disease progression during or following platinum-containing chemotherapy. Patients with EGFR or ALK genomic tumor aberrations should have disease progression on FDA-approved therapy for NSCLC harboring these aberrations prior to receiving Tecentriq.

- *Small cell lung cancer (SCLC)*

  Tecentriq, in combination with carboplatin and etoposide, is indicated for the first-line treatment of adult patients with extensive-stage small cell lung cancer (ES-SCLC).

- *Hepatocellular carcinoma (HCC)*

  Tecentriq, in combination with bevacizumab, is indicated for the treatment of patients with unresectable or metastatic HCC who have not received prior systemic therapy.

- *Melanoma*

  Tecentriq, in combination with cobimetinib and vemurafenib, is indicated for the treatment of adult patients with BRAF V600 mutation-positive unresectable or metastatic melanoma.

- *Alveolar soft part sarcoma (ASPS)*

  Tecentriq, as a single agent, is indicated for the treatment of adult and pediatric patients 2 years of age and older with unresectable or metastatic ASPS.

Compendial Uses

- Non-small cell lung cancer (NSCLC)
- Mesothelioma
- Cervical cancer
- Hepatocellular carcinoma

Atezolizumab is available as Tecentriq (Genentech, Inc.), a programmed death-ligand 1 (PD-L1) blocking antibody. Atezolizumab is a monoclonal antibody that binds to PD-L1 and blocks its interactions with both PD-1 and B7.1 receptors. This releases the PD-L1/PD-1 mediated inhibition of the

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immune response, including activation of the anti-tumor immune response without inducing antibody-dependent cellular cytotoxicity (Genentech, 2021b). Atezolizumab belongs to a class of immunotherapy

drugs known as checkpoint inhibitors, which prevent the protein PD-L1 found on some tumor cells from binding to the protein, PD-, on immune cells (Genentech, 2016). The binding of these “checkpoint” proteins suppresses the immune response, thereby allowing immune cells to attack tumors. The FDA also approved a test, called Ventana PD-L1 (sp142) to measure PD-L1 expression on patients' tumor-infiltrating immune cells. However, the test is not required for patients to receive atezolizumab, and patients whose tumors are classified as negative for PD-L1 might still respond to the therapy.

According to the prescribing information, Tecentriq carries warnings and precautions for immune-mediated adverse reactions, which may be severe or fatal, and can occur in any organ system or tissue, including the following: immune-mediated pneumonitis, immune-mediated colitis, immune-mediated hepatitis, immune-mediated endocrinopathies, immune-mediated dermatologic adverse reactions, immune-mediated nephritis and renal dysfunction, and solid organ transplant rejection; and infusion-related reactions; complications of allogeneic HSCT, of which complications can occur in patients who receive allogeneic HSCT before or after being treated with a PD-1/PD-L1 blocking antibody; and embryo-fetal toxicity, thus, the need to advise females of reproductive potential of the potential risk to a fetus and use of effective contraception.

Per the prescribing information, the most common adverse reactions (20% or more) with Tecentriq as a single-agent were fatigue/asthenia, nausea, cough, dyspnea, and decreased appetite. The most common adverse reactions (20% or more) with Tecentriq in combination with other antineoplastic drugs in patients with NSCLC and SCLC were fatigue/asthenia, nausea, alopecia, constipation, diarrhea, and decreased appetite. The most common adverse reactions (20% or more) with Tecentriq in combination with bevacizumab in patients with HCC were hypertension, fatigue and proteinuria. The most common adverse reactions (20% or more) with Tecentriq in combination with cobimetinib and vemurafenib in patients with melanoma were rash, musculoskeletal pain, fatigue, hepatotoxicity, pyrexia, nausea, pruritus, edema, stomatitis, hypothyroidism, and photosensitivity reaction.

Refer to the full prescribing information for Tecentriq for use in specific populations.

Atezolizumab was approved by the U.S. Food and Drug Administration (FDA) under an accelerated approval for metastatic triple-negative breast cancer (mTNBC; March 2019) and urothelial carcinoma. The accelerated approval for urothelial carcinoma was based on duration of response and tumor response rate. The accelerated approval for TNBC was based on progression free survival from the Phase III IMpassion130 study. The FDA approval notes that continued approval for this indication may be contingent upon verification and description of clinical benefit in confirmatory trials.

U.S. Food and Drug Administration (FDA)-Approved Indications for Atezolizumab and Hyaluronidase-tqjs (Tecentriq Hybreza)

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- Non-Small Cell Lung Cancer (NSCLC)

    - Tecentriq Hybreza, as monotherapy, is indicated as adjuvant treatment following resection and platinum-based chemotherapy for adult patients with stage II to IIIA NSCLC whose tumors have PD-L1 expression on ≥ 1% of tumor cells, as determined by an FDA-approved test.
    - Tecentriq Hybreza, as monotherapy, is indicated for the first-line treatment of adult patients with metastatic NSCLC whose tumors have high PD-L1 expression (PD-L1 stained ≥ 50% of tumor cells [TC ≥ 50%] or PD-L1 stained tumor-infiltrating immune cells [IC] covering ≥ 10% of the tumor area [IC ≥ 10%]), as determined by an FDA-approved test, with no EGFR or ALK genomic tumor aberrations.
    - Tecentriq Hybreza, in combination with bevacizumab, paclitaxel, and carboplatin, is indicated for the first-line treatment, of adult patients with metastatic non-squamous NSCLC with no EGFR or ALK genomic tumor aberrations.
    - Tecentriq Hybreza, in combination with paclitaxel protein-bound and carboplatin, is indicated for the first-line treatment of adult patients with metastatic non-squamous NSCLC with no EGFR or ALK genomic tumor aberrations.
    - Tecentriq Hybreza as monotherapy, is indicated for the treatment of adult patients with metastatic NSCLC who have disease progression during or following platinum-containing chemotherapy. Patients with EGFR or ALK genomic tumor aberrations should have disease progression on FDA-approved therapy for NSCLC harboring these aberrations prior to receiving Tecentriq Hybreza.
- Tecentriq Hybreza, in combination with carboplatin and etoposide, is indicated for the first-line treatment of adult patients with extensive-stage small cell lung cancer (ES-SCLC).
- Tecentriq Hybreza, in combination with bevacizumab, is indicated for the treatment of adult patients with unresectable or metastatic hepatocellular carcinoma (HCC) who have not received prior systemic therapy.
- Tecentriq Hybreza, in combination with cobimetinib and vemurafenib, is indicated for the treatment of adult patients with BRAF V600 mutation-positive unresectable or metastatic melanoma as determined by an FDA-approved test.
- Tecentriq Hybreza, as monotherapy, is indicated for the treatment of adult patients with unresectable or metastatic alveolar soft part sarcoma (ASPS).

Compendial Uses

- Non-small cell lung cancer
- Mesothelioma
- Hepatocellular carcinoma
- Cervical cancer
- Melanoma
- Alveolar soft part sarcoma

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Atezolizumab and hyaluronidase-tqjs is available as Tecentriq Hybreza (Genentech, Inc.), a fixed-combination drug product containing a atezolizumab and hyaluronidase (human recombinant).

- Atezolizumab is a programmed cell death ligand 1 (PD-L1) blocking antibody.
- Hyaluronidase (human recombinant) is an endoglycosidase used to increase the dispersion and absorption of co-administered drugs administered subcutaneously. It is a glycosylated single-chain protein produced by mammalian (Chinese Hamster Ovary) cells containing a DNA plasmid encoding for a soluble fragment of human hyaluronidase (PH20).

Atezolizumab is a monoclonal antibody that binds to PD-L1 and blocks its interactions with both PD-1 and B7.1 receptors. The resultant effect is PD-L1/PD-1 mediated inhibition of the immune response, including activation of the anti-tumor immune response without inducing antibodydependent cellular cytotoxicity.

Hyaluronidase increases permeability of the subcutaneous tissue by depolymerizing hyaluronan. In the doses administered, hyaluronidase in Tecentriq Hybreza works transiently and locally. Hyaluronidase effects are reversible and permeability of the subcutaneous tissue is resumes within 24 to 48 hours (Genentech, 2024b).

According to the prescribing information, Tecentriq Hybreza is contraindicated in patients with known hypersensitivity to hyaluronidase on any of its excipients.

Tecentriq Hybreza carries the following warnings and precautions:

- Immune-mediated adverse reactions:
    - Immune-mediated adverse reactions, which may be severe or fatal, can occur in any organ system or tissue, including the following: immune-mediated pneumonitis, immune-mediated colitis, immune-mediated hepatitis, immune-mediated endocrinopathies, immune-mediated dermatologic adverse reactions, immune-mediated nephritis and renal dysfunction, and solid organ transplant rejection.
    - Monitor for early identification and management. Evaluate liver enzymes, creatinine, and thyroid function at baseline and periodically during treatment.
    - Withhold or permanently discontinue based on severity and type of reaction.
- Infusion-related reactions: Pause or slow the rate of injection, or permanently discontinue based on severity of the reaction.
- Complications of allogeneic hematopoietic stem cell transplantation (HSCT): Fatal and other serious complications can occur in patients who receive allogeneic HSCT before or after being treated with a PD-1/PD-L1 blocking antibody. Follow patients closely for evidence of transplant-related complications and intervene promptly.
- Embryo-fetal toxicity: Can cause fetal harm. Advise females of reproductive potential of the potential risk to a fetus and use of effective contraception.

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The most common adverse reactions (AR) (≥ 10%) with Tecentriq Hybreza as monotherapy in patients with NSCLC were fatigue, musculoskeletal pain, cough, dyspnea, and decreased appetite.
The safety profile of Tecentriq Hybreza for treatment of NSCLC, EC-SCLC, HCC, melanoma and ASPS indications was based on safety of intravenous atezolizumab in these populations. The most common AR with intravenous atezolizumab are noted below by indication and regimen.

- Most common AR (≥ 20%) as monotherapy were:
    - First-line NSCLC: fatigue/asthenia.
    - Metastatic NSCLC: fatigue/asthenia, cough, decreased appetite, dyspnea, and myalgia/pain.
    - ASPS: musculoskeletal pain, fatigue, rash, cough, headache, nausea, hypertension, vomiting, constipation, dyspnea, dizziness, hemorrhage, diarrhea, insomnia, abdominal pain hypothyroidism, pyrexia, anxiety, arrhythmia and decreased appetite.
- Most common AR (≥ 20%) in combination with other antineoplastic drugs were:
    - NSCLC (with bevacizumab, paclitaxel, and carboplatin): neuropathy, fatigue/asthenia, alopecia, myalgia, nausea, diarrhea, constipation, decreased appetite, arthralgia, hypertension, rash, cough.
    - Non-squamous NSCLC (with paclitaxel protein-bound and carboplatin): fatigue/asthenia, nausea, diarrhea, myalgia/pain, constipation, neuropathy, alopecia, dyspnea, decreased appetite, cough, vomiting and rash.

On September 12, 2024, the U.S. Food and Drug Administration (FDA) approved atezolizumab and hyaluronidase-tqjs (Tecentriq Hybreza) given as subcutaneous injection for the same indications as the intravenous formulation, atezilozumab (Tecentriq). The FDA-approved indications for both products include non-small cell lung cancer (NSCLC), small cell lung cancer (SCLC), hepatocellular carcinoma (HCC), melanoma, and alveolar soft part sarcoma (ASPS). The FDA approval was based on supporting data from the IMscin001 study and other adequate and well-controlled studies of intravenous atezolizumab for these same indications (FDA, 2024).

The IMscin001 study design was an open-label, multi-center, international, randomized trial. Investigators evaluated the efficacy and safety of atezolizumab and hyaluronidase-tqjs (Tecentriq Hybreza) in adult patients with locally advanced or metastatic NSCLC who were not previously exposed to cancer immunotherapy and whose disease progressed following treatment with platinum-based chemotherapy. A total of 371 patients were randomized to receive either subcutaneous Tecentriq Hybreza (n = 247) or intravenous atezolizumab (n = 124). Tecentriq Hybreza was administered as 1,875 mg of atezolizumab and 30,000 units of hyaluronidase subcutaneously in the thigh every 3 weeks or atezolizumab was administered was intravenously as 1,200 mg every 3 weeks until disease progression or unacceptable toxicity (Genentech, 2024b; FDA, 2024).

The primary outcome measure was atezolizumab exposure (C_{trough} and $AUC_{0\text{-}21\ days}$) of subcutaneous Tecentriq Hybreza as compared to intravenous atezolizumab. Additional descriptive efficacy outcome

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Tecentriq Hybreza as compared to intravenous atezolizumab. Additional descriptive efficacy outcome measures were overall response rate (ORR), progression-free survival (PFS) and overall survival (OS) (Genentech, 2024b; FDA, 2024).
At the primary analysis, the geometric mean ratio (GMR) (90% CI) of subcutaneous atezolizumab and intravenous atezolizumab for Cycle 1 $_{Ctrough}$ was 1.05 (0.88, 1.24) and $AUC_{0\text{-}21days}$ was 0.87 (0.83, 0.92). This met the lower limit of the GMR (90% CI) above the pre-specified threshold of 0.8 for comparability. The confirmed ORR was 9% (95% CI: 5, 13) in the subcutaneous Tecentriq Hybreza arm and 8% (95% CI: 4, 14) in the intravenous atezolizumab arm. No notable differences in ORR, PFS or OS were observed between the different formulations (Genentech, 2024b; FDA, 2024).

The most common adverse reactions of any grade (≥ 10%) were fatigue, musculoskeletal pain, cough, dyspnea, and decreased appetite (Genentech, 2024b).

The efficacy and safety of Tecentriq Hybreza for its FDA-approved indications have been established in adequate and well-controlled studies of intravenous atezolizumab for these same indications (Genentech, 2024b).

Alveolar Soft Part Sarcoma

On December 9, 2022, the U.S. Food and Drug Administration (FDA) approved atezolizumab (Tecentriq) for adult and pediatric patients 2 years of age and older with unresectable or metastatic alveolar soft part sarcoma (ASPS). The FDA approval was based on supporting data from Study ML39345, an open-label, single-arm study consisting of 49 adult and pediatric patients with unresectable or metastatic ASPS. Patient eligibility was based on histologically or cytologically confirmed ASPS incurable by surgery and Eastern Cooperative Oncology Group (ECOG) performance status of ≤ 2. Tecentriq was administered to adult patients as 1200 mg intravenously and 15 mg/kg (up to a maximum of 1200 mg) intravenously once every 21 days until disease progression or unacceptable toxicity. The main efficacy outcome measures were overall response rate (ORR) and duration of response (DOR). The ORR was reported as 24% (95% confidence interval [CI]: 13, 39). Of the 12 patients who experienced an objective response, 67% had a DOR of 6 months or more, and 42% had a DOR of 12 months or more.

Emens (2017) noted that immunotherapy is revolutionizing the management of multiple solid tumors, and early data have revealed the clinical activity of the programmed death 1 (PD-1) or its ligand (PD-L1) antagonists in small numbers of metastatic breast cancer patients. Clinical activity appeared more likely if the tumor is triple negative, PD-L1+, and/or harbors higher levels of TILs. Responses to atezolizumab and pembrolizumab appeared to be durable in metastatic triple negative breast cancer (TNBC), suggesting these agents may transform the lives of responding patients. Current clinical efforts are focused on developing immunotherapy combinations that convert non-responders to responders, deepen those responses that do occur, and surmount acquired resistance to immunotherapy. Identifying biomarkers that can predict the potential for response to single-agent immunotherapy, identify the best immunotherapy combinations for a particular patient, and guide salvage immunotherapy in patients with

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progressive disease are high priorities for clinical development. Smart clinical trials testing rational

immunotherapy combinations that include robust biomarker evaluations will accelerate clinical progress, moving researchers closer to effective immunotherapy for almost all breast cancer patients.

On March 8, 2019, the Food and Drug Administration (FDA) granted accelerated approval for atezolizumab to be used in combination with paclitaxel protein-bound for adult patient with unresectable locally advanced or metastatic triple-negative breast cancer (TNBC) whose tumors express PD-L1 (PD-L1 stained tumor-infiltrating immune cells [IC] of any intensity covering ≥ 1% of the tumor area), as determined by an FDA-approved test. In addition, the FDA approved the VENTANA PD-L1 (SP142) Assay as a companion diagnostic device for selecting TNBC patients for atezolizumab. This indication was approved under FDA accelerated approval based on progression-free survival in the IMpassion130 trial (NCT02425891) (FDA, 2019).

Schmid et al. (2018) state that atezolizumab (Tecentriq) plus nab-paclitaxel (paclitaxel protein-bound, Abraxane) prolonged progression-free survival among patients with metastatic triple-negative breast cancer (TNBC) in both the intention-to-treat (ITT) population and the PD-L1-positive subgroup. The authors conducted a multicenter, international, double-blinded, placebo-controlled phase 3 trial (IMpassion130) which randomized (1:1) 902 patients with untreated metastatic TNBC to receive either atezolizumab (840 mg) intravenous (IV) or placebo infusions on days 1 and 15 of every 28-day cycle, plus paclitaxel protein-bound (100 mg/m2) IV on days 1, 8, and 15 of every 28-day cycle. Patients continued the intervention until disease progression, or an unacceptable level of toxic effects occurred. Patients were stratified by presence of liver metastases, prior taxane treatment, and by PD-L1 expression status in tumor infiltrating immune cells (PD-L1 stained tumor-infiltrating immune cells [IC] <1% of tumor area vs. ≥ 1% of the tumor area) by the VENTANA PD-L1 (SP142) Assay. The two primary end points were progression-free survival (PFS) (in the intention-to-treat (ITT) population and PD-L1-positive subgroup) and overall survival (tested in the ITT population; if the finding was significant, then it would be tested in the PD-L1-positive subgroup). Of the 902 patients in the intent to treat population (ITT), 41% (369 patients) were classified as PD-L1 expression ≥ 1%. Patients were excluded if they had a history of autoimmune disease, administration of a live attenuated vaccine within 4 weeks prior to randomization, administration of systemic immunostimulatory agents within 4 weeks or systemic immunosuppressive medications within 2 weeks prior to randomization; or untreated or corticosteroid-dependent brain metastases. The first primary end point outcomes in the ITT analysis reflected a median PFS of 7.2 months with atezolizumab plus nab-paclitaxel, as compared with 5.5 months with placebo plus nab-paclitaxel (p=0.002). Among patients with PD-L1-positive tumors, the median PFS was 7.5 months and 5.0 months, respectively (p<0.001). The outcomes of the second primary end point in the ITT analysis reflected the median overall survival of 21.3 months with atezolizumab plus nab-paclitaxel and 17.6 months with placebo plus nab-paclitaxel (p=0.08). Among patients with PD-L1-positive tumors, the median overall survival was 25.0 months and 15.5 months, respectively. No new adverse effects were identified. Adverse events that led to the discontinuation of any agent occurred in 15.9% of the patients who received atezolizumab plus nab-paclitaxel and in 8.2% of those who received placebo plus nab-paclitaxel. In conclusion, in patients with metastatic TNBC whose tumors express PD-L1

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plus nab-paclitaxel. In conclusion, in patients with metastatic TNBC whose tumors express PD-L1, study outcomes for progression-free survival were in favor of the atezolizumab plus paclitaxel protein-

bound arm; however, overall survival data were immature with 43% deaths in the intent to treat (ITT) population.

In August 2021, Genentech announced the decision to voluntarily withdraw the U.S. accelerated approval for Tecentriq in combination with chemotherapy (Abraxane®, albumin-bound paclitaxel; nab-paclitaxel) for the treatment of adults with unresectable locally advanced or metastatic triple-negative breast cancer (mTNBC) whose tumors express PD-L1, as determined by an FDA-approved test. Genentech made this decision following consultation with the FDA based on the agency's assessment of the current mTNBC treatment landscape and in accordance with the requirements of the accelerated approval program. The contingent results for the postmarketing requirement from the IMpassion131 Phase III study did not meet its primary endpoint of PFS for the initial (first-line) treatment of people with mTNBC in the PD-L1-positive population (Genentech, 2021a).

Leung et al (2022) stated that no therapeutic targets in mTNBC have been established and no effective therapies are available. Several therapies directed at novel targets and also immunotherapies have recently shown promising results in advanced or metastatic TNBC. In a meta-analysis, these researchers compared the safety and effectiveness of these new targeted and immunotherapy regimens for the treatment of mTNBC. The PubMed, Embase, and Cochrane Library electronic databases were searched for phase-III randomized trials. Trial quality was examined using the Grading of Recommendations, Assessment, Development and Evaluations (GRADE) method. The comparative outcomes were PFS, OS, and grade 3 to 4 adverse drug events (ADEs). A total of 13 phase-III randomized controlled trials (RCTs) were identified in the network meta-analysis. Olaparib significantly improved PFS in comparison with the pembrolizumab plus chemotherapy, atezolizumab plus nab-paclitaxel and pembrolizumab regimens. Sacituzumab yielded a significant improvement in OS over immunotherapies, veliparib and chemotherapy alone; but no significant superiority over pembrolizumab, olaparib and talazoparib. The risk of grade-3 or higher ADEs associated with olaparib was significantly lower than the risks associated with the other regimens. The authors concluded that for mTNBC, sacituzumab had a better effect on OS, with comparatively high risk of serious adverse events (SAE), and olaparib improved PFS with a lower risk of SAE, especially in those patients with BRCA mutations.

Atezolizumab for the Treatment of Large Cell Neuroendocrine Carcinoma

Vrontis et al (2022) noted that large cell neuroendocrine carcinoma of the lung (L-LCNEC) is a rare type of neuroendocrine lung cancer that is increasingly diagnosed; however, the optimal management regarding the advanced stage is unclear. In a retrospective study, these researchers presented their experience when L-LCNEC was treated as SCLC; a total of 8 cases of L-LCNEC were included. They reviewed medical files and reports by accessing the Institution's data of patients diagnosed with L-LCNEC from April 2019 until December 2020; and examined their response to the combination of platinum/etoposide/atezolizumab as 1st-line chemotherapy. The ORR was 75 %; he median PFS was

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6.85 months; and the median response duration was 5.5 months. The authors concluded that comparing their findings with other retrospective and prospective studies, it appeared that the systematic treatment

of choice and management in L-LCNEC of the lung should be that of a small cell carcinoma (SCC) of the lung.

Imai and Kitamura (2023) reported that a modified regimen based on IMpower 133 (carboplatin + etoposide + atezolizumab) was administered to a patient diagnosed with large cell neuroendocrine carcinoma (LCNEC) who was concurrently undergoing hemodialysis. Adverse events (AEs) resulted the discontinuation of carboplatin and etoposide following the 1st course. Nevertheless, the patient exhibited reduction in pulmonary nodule and adrenal metastasis while receiving atezolizumab, indicating its sustained effectiveness for a duration of 7 months. The authors concluded that to the best of their knowledge, this was the 1st documented case reporting successful treatment with atezolizumab in LCNEC patients undergoing hemodialysis. Atezolizumab can be administered safely in patients undergoing dialysis and is a promising therapeutic option for dialysis patients with LCNEC.

Atezolizumab for the Treatment of Urinary Tract Squamous Cell Carcinoma

Crabb et al (2023) stated that bladder and urinary tract cancers account for approximately 21,000 new diagnoses and 5,000 deaths annually in the U.K. Approximately 90 % are transitional cell carcinomas where advanced disease is treated with platinum-based chemotherapy and PD-1/PD-L1 directed immunotherapy. Urinary tract squamous cell carcinoma (UTSCC) accounts for about 5 % of urinary tract cancers overall making this a rare disease. Researchers have yet to establish definitive systemic therapeutic options for advanced UTSCC. Preliminary translational data, from UTSCC patient tumor samples, indicate high PD-L1 expression and tumor infiltrating lymphocytes in a proportion of cases. Both of these features are associated with differential gene expression consistent with a tumor/immune micro-environment predicted to be susceptible to immune checkpoint directed immunotherapy which these researchers will examine in the AURORA Trial. AURORA is an open-label, single-arm, multi-center, phase-II clinical trial. A total of 33 patients will be recruited from U.K. secondary care sites. Patients with UTSCC, suitable for treatment with palliative intent, will receive atezolizumab PD-L1 directed immunotherapy (IV infusion, 1,680 mg, every 28 days) for 1 year if tolerated. Response assessment, by cross sectional imaging will occur every 12 weeks. The AURORA Trial uses a Simon's 2-stage optimal design with best ORR (by RECIST v1.1) at a minimum of 12 weeks from commencing treatment as the primary endpoint. Secondary endpoints will include OS, PFS, DOR, magnitude of response using waterfall plots of target lesion measurements, quality of life (QOL) using the EORTC QLQ-C30 tool, safety and tolerability (CTCAE v5) and evaluation of potential biomarkers of treatment response including PD-L1 expression. Archival tumor samples and blood samples will be collected for translational analyses. The authors concluded that if this trial shows atezolizumab to be safe and effective, it may lead to a future late phase RCT in UTSCC. Ultimately, these researchers hope to provide a new therapeutic option for such patients.

Atezolizumab Plus Cobimetinib for the Treatment of Melanoma with Central Nervous System

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Metastases

Dummer et al (2023) noted that targeted therapy and immunotherapy have demonstrated intra-cranial activity in melanoma with central nervous system (CNS) metastases; however, there remains an unmet need, especially for patients with symptomatic CNS metastases. In an open-label, single-arm, multi-center, phase-II clinical trial (the TRICOTEL Trial), these researchers examined atezolizumab in combination with cobimetinib or vemurafenib plus cobimetinib in patients with melanoma with CNS metastases. This trial was carried out in 2 cohorts: a BRAFV600 wild-type cohort and a BRAFV600 mutation-positive cohort, recruited at 21 hospitals and oncology centers in Brazil, France, Germany, Hungary, Italy, Spain, and Switzerland. Eligible patients were aged 18 years or older with previously untreated metastatic melanoma, brain metastases of 5 mm or larger in at least 1 dimension, and an ECOG performance status (PS) of 2 or less. Patients in the BRAFV600 wild-type cohort received IV atezolizumab (840 mg, days 1 and 15 of each 28-day cycle) plus oral cobimetinib (60 mg once-daily, days 1 to 21). Patients in the BRAFV600 mutation-positive cohort received IV atezolizumab (840 mg, days 1 and 15 of each 28-day cycle) plus oral vemurafenib (720 mg twice-daily) plus oral cobimetinib (60 mg once-daily, days 1 to 21); atezolizumab was withheld in cycle 1. Treatment was continued until progression, toxicity, or death. The primary outcome was intra-cranial ORR confirmed by assessments at least 4 weeks apart, as evaluated by IRC using modified RECIST version 1.1. Because of early closure of the BRAFV600 wild-type cohort, the primary endpoint of intra-cranial ORR by IRC assessment was not carried out in this cohort; intra-cranial ORR by investigator assessment was reported instead. Safety and effectiveness were analyzed in all patients who received at least 1 dose of study medication. Between December 13, 2018, and December 7, 2020, a total of 65 patients were enrolled in the BRAFV600 mutation-positive cohort; the BRAFV600 wild-type cohort was closed early after enrolment of 15 patients. Median follow-up was 9.7 months (IQR 6.3 to 15.0) for the BRAFV600 mutation-positive cohort and 6.2 months (3.5 to 23.0) for the BRAFV600 wild-type cohort. Intra-cranial ORR was 42 % (95 % CI: 29 to 54) by IRC assessment in the BRAFV600 mutation-positive cohort and 27 % (95 % CI: 8 to 55) by investigator assessment in the BRAFV600 wild-type cohort. Treatment-related grade-3 or worse AEs occurred in 41 (68 %) of 60 patients who received atezolizumab plus vemurafenib plus cobimetinib in the BRAFV600 mutation-positive cohort, the most common of which were lipase increased (15 [25 %] of 60 patients) and blood creatine phosphokinase increased (11 [18 %]); 8 (53 %) of 15 patients treated with atezolizumab plus cobimetinib in the BRAFV600 wild-type cohort had treatment-related grade-3 or worse AEs, most commonly anemia (2 [13 %]) and dermatitis acneiform (2 [13 %]) . Treatment-related serious AEs occurred in 14 (23 %) of 60 patients who received triplet therapy in the BRAFV600 mutation-positive cohort and 2 (13 %) of 15 in the BRAFV600 wild-type cohort. No treatment-related deaths occurred. The authors concluded that atezolizumab plus cobimetinib provided intra-cranial activity in patients with BRAFV600-mutated melanoma with CNS metastases.

Atezolizumab, With or Without Radiotherapy, for the Treatment of Squamous Cell Carcinoma of the Penis

de Vries et al (2023) noted that patients with advanced squamous cell carcinoma (SCC) of the penis have

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a poor prognosis (21 % 2-year OS from diagnosis). In a non-randomized, single-center, phase-II clinical trial, these researchers examined the activity of atezolizumab (anti-PD-L1) in patients with advanced

penile cancer, with or without radiotherapy (RT). This study had 2 treatment arms totaling 32 patients with histologically confirmed advanced penile cancer. All participants received atezolizumab (1,200 mg) once every 3 weeks; 20 patients, who were expected to benefit from RT for loco-regional disease control, received additional irradiation. The primary endpoint was 1-year PFS for the complete cohort and was reached if the actual 1-year PFS was at least 35 %. Secondary endpoints included OS, ORR, and tolerability. Exploratory biomarker analyses were carried out in pre-treatment specimens. Median follow-up was 29.1 months (inter-quartile range [IQR]: 18.1 to 33.5); grade-3 to grade-4 AEs related to atezolizumab or RT were observed in 3/32 (9.4 %) and 13/20 (65%) patients, respectively. 1-year PFS was 12.5 % (95 % CI: 5.0 to 31.3), which did not meet the study's primary endpoint. Median OS was 11.3 months (95 % CI: 5.5 to 18.7). In the objective response-evaluable population (n = 30; 93.8 %), the ORR was 16.7 % (95 % CI: 6 % to 35 %), including 2 (6.7 %) complete responders (CRs) and 3 (10 %) partial responders (PRs). Improved PFS was observed in patients with high-risk human papillomavirus (hrHPV)-positive tumors (p = 0.003) and those with high infiltration of intra-tumoral CD3+CD8+ T cells (p = 0.037). The authors concluded that although the primary endpoint of 1-year PFS was not met, durable anti-tumor activity to atezolizumab was observed in a subset of patients. Biomarkers, such as hrHPV and intra-tumoral CD3+CD8+ T-cell infiltration, may aid in better selecting responders.

Combined Atezolizumab, Bevacizumab, and Radiation Therapy for the Treatment of Hepatocellular Carcinoma

Kim et al (2023) stated that radiotherapy (RT) is an effective local treatment for the treatment of HCC; however, whether additional RT is safe and effective in patients with advanced HCC receiving atezolizumab plus bevacizumab remains unclear. In a retrospective, cohort study, these researchers examined the feasibility of additional RT in these patients. They analyzed 7 patients with advanced HCC who received RT during treatment with atezolizumab plus bevacizumab. The median prescribed RT dose was 35 Gy (range of 33 to 66). Freedom from local progression (FFLP), PFS, and OS following RT were analyzed. The median follow-up duration following RT was 14.2 months (range of 10.0 to 18.6). Of the 7 patients, disease progression was noted in 6 (85.7 %), the sites of disease progression were local in 2 (28.6 %), intra-hepatic in 4 (57.1 %), and extra-hepatic in 4 (57.1 %). The median time of FFLP was not reached, and PFS and OS times were 4.0 (95 % CI: 3.6 to 4.5) and 14.8 % (95 % CI: 12.5 to 17.2) months, respectively. The 1-year FFLP, PFS, and OS rates were 60 % (95 % CI: 43.8 % to 76.2 %), 0 %, and 85.7 % (95 % CI: 75.9 % to 95.5 %), respectively. Grade-3 or higher hematologic AEs were not observed, but grade-3 non-hematologic AEs unrelated to RT were observed in 1 patient. The authors concluded that the addition of RT may be feasible in patients with advanced HCC treated with atezolizumab plus bevacizumab; however, further prospective, large-scale studies are needed to validate these findings, and evaluate the safety and effectiveness of this therapeutic approach.

Combined Atezolizumab and Bevacizumab for the Treatment of Hepatocellular Carcinoma

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Xin et al (2022) noted that atezolizumab plus bevacizumab has been shown to have promising anti-tumor activity and tolerable safety in patients with unresectable HCC. Hepatic arterial infusion

chemotherapy (HAIC) has also been shown to exhibit high response rates and favorable survival for patients with advanced HCC. These researchers examined the safety and effectiveness of atezolizumab plus bevacizumab combined with HAIC for patients with treatment-naive advanced HCC. Between October 2020 and September 2021, patients with advanced HCC who initially received atezolizumab plus bevacizumab combined with HAIC of oxaliplatin, fluorouracil, and leucovorin (FOLFOX) from 3 hospitals in China were reviewed for eligibility. The effectiveness was examined by tumor response rate and survival, and the safety was examined by the frequency of main AEs. A total of 52 eligible patients with advanced HCC who received triple therapy were included in this study. The ORRs based on modified RECIST and RECIST1.1 criteria were 67.3 % and 44.2 %, respectively. The median PFS was 10.6 months (95 % CI: 8.37 to 13.8), and OS was not reached. Extra-hepatic metastasis was an independent risk factor associated with PFS. All AEs were managed, and no treatment-related deaths occurred. The authors concluded that atezolizumab plus bevacizumab combined with HAIC-FOLFOX had a significant therapeutic effect and manageable AEs in patients with advanced HCC, which may be a potential therapeutic option for advanced HCC.

Combined Tiragolumab and Atezolizumab for the Treatment of Solid Tumors

In a non-randomized controlled, phase-Ia/Ib trial, Kim et al (2023) examined the safety and anti-tumor activity of tiragolumab and its combination with atezolizumab in patients with advanced solid tumors. This study was carried out at 13 sites in 6 countries (Australia, Canada, France, Korea, Spain, and the U.S.). The start dates were May 23, 2016, for phase-Ia and October 11, 2016, for phase-Ib. Patients were aged 18 years or older with measurable disease at baseline. The clinical cut-off date was October 1, 2021. Data analysis was carried out on January 24, 2022. Patients received fixed-dose IV tiragolumab on day 1 of each 21-day cycle (2 mg escalating to 1,200 mg) in phase-Ia, plus fixed-dose IV atezolizumab (1,200 mg every 3 weeks) in phase-Ib. Patients were treated until disease progression, loss of clinical benefit, or development of unacceptable toxicity. The primary endpoints included the safety, tolerability, and recommended phase-II dose (RP2D) of tiragolumab or combination tiragolumab plus atezolizumab. The secondary endpoint included the investigator-assessed ORR. Counts and percentages were used for categorical variables, and medians and ranges were used for continuous variables. Among the phase-Ia (n = 24) and 1b (n = 49) dose-escalation cohorts, the median age was 60 (range of 40 to 77) and 54 (range of 25 to 81) years, respectively. More than 50 % of patients were women (14 of 24 [58 %] and 25 of 49 [51 %]), and more than 1/3 (10 [42 %] and 18 [37 %]) had received 4 or more prior cancer therapies. No dose-limiting toxicities (DLTs) occurred, and the maximum tolerated dose (MTD) of tiragolumab was not reached (NR). The most frequent treatment-related adverse events (TRAEs) were fatigue (5 of 24 [21 %]) in phase-Ia and pruritus (5 of 49 [10 %]) in phase-Ib; the majority of AEs were grade-1 or grade-2. Immune-mediated AEs occurred in 4 of 24 (17 %) and 29 of 49 (59 %) patients during phases-Ia and phase-Ib, respectively (primarily grade-1 or grade-2). The RP2D of tiragolumab was 600 mg intravenously every 3 weeks, which was tested in phase-Ib dose expansion. The confirmed ORR was 0 % during phase-Ia, with evidence of anti-tumor activity in 6 % of patients (n = 3) during

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phase-Ib. The safety profile of combination tiragolumab plus atezolizumab in phase-Ib was similar in the dose-escalation and dose-expansion cohorts. The confirmed ORR was 46 % (6 of 13) in the NSCLC

cohort (median DOR, NR) and 28 % (5 of 18) in the esophageal cancer (EC) cohort (median DOR, 15.2 [95 % CI: 7.0 to NR] months). The authors concluded that in this non-randomized controlled trial, tiragolumab was well-tolerated with or without atezolizumab; no new safety signals were observed. These researchers stated that preliminary anti-tumor activity was showed for the combination regimen in patients with cancer immunotherapy-naive metastatic NSCLC or EC.

Gastrointestinal Cancers (e.g. Colorectal, Esophageal and Gastric Cancers)

Basile and co-workers (2017) stated that in the past few years, significant advances in molecular biology have provided new therapeutic options for colorectal cancer (CRC). The development of new drugs that target the immune response to cancer cells appeared very promising and has already been established for other tumor types. In particular, the use of immune checkpoint inhibitors appeared to be an encouraging immunotherapeutic strategy. These investigators provided an update of the current evidence related to this topic, though most immunotherapies are still in early-phase clinical trials for CRC. To understand the key role of immunotherapy in CRC, these researchers discussed the delicate balance between immune-stimulating and immune-suppressive networks that occur in the tumor microenvironment. Modulation of the immune system through checkpoint inhibition is an emerging approach in CRC therapy. Nevertheless, selection criteria that could enable the identification of patients who may benefit from these agents are necessary. Furthermore, potential prognostic and predictive immune biomarkers based on immune and molecular classifications have been proposed. The authors concluded that additional studies are needed to develop biomarkers, effective therapeutic strategies and novel combinations to overcome immune escape resistance and enhance effector response; atezolizumab was one of the keywords of this article.

Bilgin and colleagues (2017) stated that VEGF, HER2 and EGFR targeted agents are currently used in gastric, esophageal and colorectal cancers. However, treatment outcomes are still poor in most gastro-intestinal (GI) cancers. Immune checkpoints are one of the most promising immunotherapy approaches. These investigators discussed the safety and effectiveness of anti-PD-1/PD-L1 therapies in GI cancers, including gastric, esophageal and colorectal cancer in published or reported studies. They performed a literature search from PubMed and ASCO Annual Meeting abstracts by using the following search keywords: "nivolumab", "pembrolizumab", "avelumab", "GI cancers" "anti-PD1 therapy" and "anti-PD-L1 therapy". The last search was on November 2, 2016. The most important limitation of this review was that most of the data on anti-PD-1/PD-L1 therapies in GI cancers relied on phase-I and phase-II clinical trials. Currently, there are 2 anti-PD-1 (nivolumab and pembrolizumab) and 1 anti-PDL1 (atezolizumab) agents approved by FDA. After the treatment efficacy of immune checkpoint blockade was shown in melanoma, renal cell cancer (RCC) and non-squamous lung cancer, trials that evaluated immune checkpoint blockade in GI cancers are ongoing. Early results of trials have been promising and encouraging for patients with advanced stage gastro-esophageal cancer. According to early results of published trials, response to anti-PD1/PD-L1 agents appeared to be associated with

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tumor PD-L1 levels. According to 2 published phase-II clinical trials, the clinical benefits of immune checkpoint blockade with both nivolumab and pembrolizumab were limited in patients with

microsatellite instability (MSI)-positive advanced CRC. However, several phase II/III clinical trials are still ongoing. The authors concluded that both pembrolizumab and nivolumab showed promising efficacy with acceptable safety data in published trials in GI cancers, especially in refractory MSI positive metastatic CRC.

Melanoma

Grimaldi and associates (2017) stated that the mitogen-activated protein kinase (MAPK) cascade is an intracellular signaling pathway involved in the regulation of cellular proliferation and the survival of tumor cells. Several different mutations, involving BRAF or NRAS, exert an oncogenic effect by activating the MAPK pathway, resulting in an increase in cellular proliferation. These mutations have become targets for new therapeutic strategies in melanoma and other cancers. Selective MEK inhibitors have the ability to inhibit growth and induce cell death in BRAF- and NRAS-mutant melanoma cell lines. MEK inhibitor therapy in combination with a BRAF inhibitor is more effective and less toxic than treatment with a BRAF inhibitor alone, and has become the standard of care for patients with BRAF-mutated melanoma. Trametinib was the 1st MEK inhibitor approved for the treatment of BRAF-mutated metastatic melanoma not previously treated with BRAF inhibitors, and is also approved in combination with the BRAF inhibitor dabrafenib. Furthermore, cobimetinib is another MEK inhibitor approved for the treatment of BRAF-mutated metastatic melanoma in combination with a BRAF inhibitor, vemurafenib. The MEK inhibitor binimetinib in combination with the BRAF inhibitor encorafenib is in clinical development. The addition of an anti-PD-1/PD-L1 agent (e.g., pembrolizumab, durvalumab or atezolizumab) to combined BRAF and MEK inhibition has shown considerable promise, with several trials ongoing in metastatic melanoma. Binimetinib has also shown efficacy in NRAS-mutated melanoma patients. The authors stated that future possibilities for MEK inhibitors in advanced melanoma, as well as other solid tumors, include their use in combination with other targeted therapies (e.g. anti-CDK4/6 inhibitors) and/or various immune-modulating antibodies.

In July 2020, the FDA approved atezolizumab (Tecentriq, Genentech, Inc.) in combination with cobimetinib and vemurafenib for patients with BRAF V600 mutation-positive unresectable or metastatic melanoma. FDA approval was based on results from the double-blind, randomized (1:1), placebo-controlled, multicenter (IMspire150; NCT02908672) study. which found that the addition of atezolizumab to targeted therapy with vemurafenib and cobimetinib was safe and tolerable and significantly increased progression-free survival in patients with BRAFV600 mutation-positive advanced melanoma (FDA, 2020; Gutzmer et al., 2020).

In the IMspire150 trial, 514 patients with unresectable stage IIIc-IV, BRAFV600 mutation-positive melanoma were randomly assigned 1:1 to 28-day cycles of atezolizumab, vemurafenib, and cobimetinib (atezolizumab group) or atezolizumab placebo, vemurafenib, and cobimetinib (control group). In cycle 1, all patients received vemurafenib and cobimetinib only; atezolizumab placebo was added from cycle

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2 onward. Randomization was stratified by lactate dehydrogenase concentration and geographical region. Blinding for atezolizumab was achieved by means of an identical intravenous placebo, and

blinding for vemurafenib was achieved by means of a placebo tablet. The primary outcome was investigator-assessed progression-free survival. At a median follow-up of 18.9 months, progression-free survival as assessed by the study investigator was significantly prolonged with atezolizumab versus control (15.1 vs 10.6 months; p=0·025). Common treatment-related adverse events (>30%) in the atezolizumab and control groups were blood creatinine phosphokinase increased (51.3% vs 44.8%), diarrhea (42.2% vs 46.6%), rash (40.9%, both groups), arthralgia (39.1% vs 28.1%), pyrexia (38.7% vs 26.0%), alanine aminotransferase increased (33.9% vs 22.8%), and lipase increased (32.2% vs 27·.%); 13% of patients in the atezolizumab group and 16% in the control group stopped all treatment because of adverse events (Gutzmer et al., 2020). In conclusion, the primary efficacy outcome measure was investigator-assessed progression-free survival (PFS) per RECIST 1.1. Median PFS was 15.1 months in the atezolizumab arm and 10.6 months in the placebo arm (p=0.0249) (FDA, 2020).

Melanoma with CNS Metastases

Dummer et al (2022) noted that targeted therapy and immunotherapy have shown intra-cranial activity in melanoma with CNS metastases; however, there remains an unmet need, especially for patients with symptomatic CNS metastases. In an open-label, single-arm, multi-center, phase-II clinical trial (the TRICOTEL Trial), these researchers examined atezolizumab in combination with cobimetinib or vemurafenib plus cobimetinib in patients with melanoma with CNS metastases.

This trial was carried out in 2 cohorts: a BRAFV600 wild-type cohort and a BRAFV600 mutation-positive cohort; subjects were recruited at 21 hospitals and oncology centers in Brazil, France, Germany, Hungary, Italy, Spain, and Switzerland. Eligible patients were 18 years of age or older with previously untreated metastatic melanoma, CNS metastases of 5-mm or larger in at least 1 dimension, and an Eastern Cooperative Oncology Group performance status (ECOG PS) of 2 or less. Patients in the BRAFV600 wild-type cohort received intravenous (IV) atezolizumab (840 mg, days 1 and 15 of each 28-day cycle) plus oral cobimetinib (60 mg once-daily, days 1 to 21). Patients in the BRAFV600 mutation-positive cohort received IV atezolizumab (840 mg, days 1 and 15 of each 28-day cycle) plus oral vemurafenib (720 mg twice-daily) plus oral cobimetinib (60 mg once-daily, days 1 to 21); atezolizumab was withheld in cycle 1. Treatment was continued until progression, toxicity, or death. The primary outcome was intra-cranial ORR confirmed by assessments at least 4 weeks apart, as assessed by independent review committee (IRC) using modified RECIST version 1.1. Because of early closure of the BRAFV600 wild-type cohort, the primary endpoint of intra-cranial ORR by IRC assessment was not performed in this cohort; intra-cranial ORR by investigator assessment was reported instead. Safety and effectiveness were analyzed in all patients who received at least 1 dose of study medication. Between December 13, 2018, and December 7, 2020, a total of 65 patients were enrolled in the BRAFV600 mutation-positive cohort; the BRAFV600 wild-type cohort was closed early after enrolment of 15 patients. Median follow-up was 9.7 months (inter-quartile range [IQR] 6.3 to 15.0) for the BRAFV600 mutation-positive cohort and 6.2 months (3.5 to 23.0) for the BRAFV600 wild-type

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cohort. Intra-cranial ORR was 42 % (95 % CI: 29 to 54) by IRC assessment in the BRAFV600 mutation-positive cohort and 27 % (95 % CI: 8 to 55) by investigator assessment in the BRAFV600

wild-type cohort. Treatment-related grade-3 or worse AEs occurred in 41 (68 %) of 60 patients who received atezolizumab plus vemurafenib plus cobimetinib in the BRAFV600 mutation-positive cohort, the most common of which were lipase increased (15 [25 %] of 60 patients) and blood creatine phosphokinase increased (10 [17 %]); 8 (53 %) of 15 patients treated with atezolizumab plus cobimetinib in the BRAFV600 wild-type cohort had treatment-related grade-3 or worse AEs, most commonly anemia (2 [13 %]) and dermatitis acneiform (2 [13 %]). Treatment-related SAEs occurred in 14 (23 %) of 60 patients in the BRAFV600 mutation-positive cohort and 2 (13 %) of 15 in the BRAFV600 wild-type cohort; 1 death in the BRAFV600 mutation-positive cohort (limbic encephalitis) was considered to be related to atezolizumab treatment. The authors concluded that the addition of atezolizumab to vemurafenib plus cobimetinib provided promising intra-cranial activity in patients with BRAFV600-mutated melanoma with CNS metastases.

Metastatic Neuroendocrine Tumor of the Intestines

Schmidt and Wiedenmann (2018) noted that treatment and prognosis of neuroendocrine neoplasia depends on tumor size, stage, grade, resectability, and extent of distant metastasis. In most cases a multi-modality approach including surgical, locally invasive procedures, peptide-guided radioreceptor therapy (PRRT), and medical therapies represent the mainstay of treatment in advanced disease. In the reported case, a 68-year-old man was diagnosed in 2010 with an initially functional (histamine) neuroendocrine tumor of gastric type III, G2, stage IVB, cT4cN1cM1 (hepatic, peritoneal, nodal, osseous), including a hepatic tumor load of 25 %. Intensive multi-modality approaches including combined immunotherapy (vaccination and PD-1/CTLA-4 blockade) led to a survival of 8 years until now with a high quality of life (QOL) and minimal residual disease (MRD; only a single, small para-gastric recurrence) despite the dedifferentiation of the tumor into a neuroendocrine carcinoma G3 (Ki-67 of 80 %) including a non-functional stage. The authors concluded that based on the fact that angiogenesis and immunosuppression frequently occur simultaneously in response to diverse stimuli, they considered, aside from the combination of conventional treatment strategies (TACE, AL, PRRT, HIPEC, CTX, and surgery), the use of immunotherapy as a future option for the treatment of at least a subgroup of metastatic NET patients (atezolizumab was one of the chemotherapeutic agents used in this study).

Wee et al (2021) stated that chemotherapy in combination with immunotherapy has a proven survival benefit compared to chemotherapy alone in extensive stage SCLC and is the new standard of care (SOC). Since extra-pulmonary small cell carcinomas (SCCs) are less common, treatment paradigms are reasonably extrapolated from SCLC regimens. These researchers examined their experience on the use of combined chemotherapy and immunotherapy (as used in SCLC) for SCC and neuroendocrine carcinoma of the prostate. Using an institutional database search tool, these researchers searched the electronic medical record to identify patients with SCC or neuroendocrine carcinoma of the prostate who had been treated with atezolizumab and chemotherapy. They recorded patient characteristics, including

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age, pathology, and disease extent. Treatment characteristics included number of prior treatments, use of concomitant androgen deprivation, number of cycles of immunotherapy, and prior systemic therapies

(including those for adenocarcinoma of the prostate); PFS and OS were the primary outcomes. These investigators identified 7 men who received atezolizumab for metastatic prostate cancer with a small cell or neuroendocrine component. In 6 of the 7 patients, the combination of carboplatin, etoposide, and atezolizumab was the 1st-line of treatment after diagnosis of small cell or neuroendocrine carcinoma; 2 of the 7 patients had de-novo small cell/neuroendocrine pathology, while the other 5 had transformation from a pre-existing adenocarcinoma. In the patients who received chemotherapy plus immunotherapy in the 1st-line setting, at a median follow-up of 6.5 months (range of 1.5 to 15.1), the median PFS was 3.4 months, and median OS was 8.4 months. The authors concluded that small cell or neuroendocrine carcinoma of the prostate was associated with poor survival outcomes despite adding immunotherapy (atezolizumab) to chemotherapy (carboplatin and etoposide). These researchers stated that there has been no demonstrable benefit of adding immunotherapy to chemotherapy in this setting.

Halperin et al (2022) stated that therapies for patients with advanced well-differentiated neuroendocrine tumors (NETs) have expanded but remain inadequate, with patients dying of disease despite recent advances in NET therapy. While patients with other cancers have seen long-term disease control and tumor regression with the use of immunotherapies, initial prospective studies of single-agent programmed cell death 1 (PCD-1) inhibitors in NET have been disappointing. In a single-arm, open-label, non-randomized clinical trial, these researchers examined the response rate following treatment with the combination of the VEGF inhibitor bevacizumab with the PCD-1 ligand 1 inhibitor atezolizumab in patients with advanced NETs. This study enrolled patients with rare cancers including 40 subjects with advanced, progressive grade-1 to grade-2 NETs (20 with pancreatic NETs [pNETs] and 20 with extra-pancreatic NETs [epNETs]) treated at a tertiary care referral cancer center between March 31, 2017, and February 19, 2019. Data were analyzed from June to September 2021. Patients received IV bevacizumab and atezolizumab at standard doses every 3 weeks until progression, death, or withdrawal. The primary endpoint was objective radiographic response using RECIST, version 1.1, with PFS as a key secondary endpoint. Following treatment of the 40 study patients with bevacizumab and atezolizumab, objective response was observed in 4 patients with pNETs (20 %; 95 % CI: 5.7 % to 43.7 %) and 3 patients with epNETs (15 %; 95 % CI: 3.2 % to 37.9 %). The PFS was 14.9 (95 % CI: 4.4 to 32.0) months and 14.2 (95 % CI: 10.2 to 19.6) months in these cohorts, respectively. The authors concluded that these findings suggested that clinical responses in patients with NET may follow treatment with the combination of bevacizumab and atezolizumab, with a PFS consistent with effective therapies.

Metastatic Non-Squamous Non-Small Cell Lung Cancer (NSCLC)

On December 4, 2019, the U.S. FDA approved Tecentriq (atezolizumab) in combination with chemotherapy (Abraxane [paclitaxel protein-bound; nab-paclitaxel] and carboplatin) for the initial (first-line) treatment of adults with metastatic non-squamous non-small cell lung cancer (NSCLC) with no EGFR or ALK genomic tumour aberrations. FDA approval is based on the safety and efficacy from a

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multicenter, international, randomized, open-label, Phase 3, IMpower130 study in which chemotherapy-naïve patients with metastatic non-squamous NSCLC received atezolizumab 1200 mg and carboplatin

AUC 6 mg/mL/min intravenously on Day 1 and paclitaxel protein-bound 100 mg/m^2 intravenously on Day 1, 8, and 15 of each 21-day cycle for a maximum of 4 or 6 cycles, followed by atezolizumab 1200 mg intravenously every 3 weeks until disease progression or unacceptability toxicity. Patients could have received prior EGFR or ALK kinase inhibitor, if appropriate. Among patients receiving atezolizumab (Tecentriq), 55% were exposed for 6 months or longer and 3.5% were exposed for greater than one year. The outcomes of the study showed that atezolizumab plus chemotherapy demonstrated a significant overall survival (OS) compared to chemotherapy alone (median [OS]=18.6 versus 13.9 months; p=0.0384) in the intention-to-treat wild-type (ITT-WT) population, as well as, significantly improved progression-free survival (PFS) compared with chemotherapy alone (median PFS=7.2 versus 6.5 months; p=0.0024) in the ITT-WT population. PFS and OS benefit was observed in all PD-L1 subgroups, and consistently across all subgroups, except in patients with liver metastases and EGFR/ALK genomic alterations. (Capuzzo et al, 2018; Genentech, 2019; Roche, 2019).

Safety for the atezolizumab plus chemotherapy combination appeared consistent with the known safety profiles of the individual medicines, and no new safety signals were identified with the combination. Grade 3-4 treatment-related adverse events were reported in 73.2% of people receiving atezolizumab plus chemotherapy compared with 60.3% of those receiving chemotherapy alone (Roche, 2019).

Non-Small Cell Lung Cancer (NSCLC)

The FDA approved atezolizumab for the treatment of people with metastatic non-small cell lung cancer (NSCLC) who have disease progression during or following platinum-containing chemotherapy and have progressed on an appropriate FDA-approved targeted therapy if their tumor has EGFR or ALK gene abnormalities.

The approval was based on results from the randomized Phase III OAK and Phase II POPLAR studies. OAK was a multicenter, open-label, randomized, controlled Phase III study that evaluated the efficacy and safety of atezolizumab compared with docetaxel in 1,225 subjects with locally advanced or metastatic NSCLC whose disease had progressed following previous treatment with platinum-containing chemotherapy, with the primary analysis consisting of the first 850 randomized patients. The study enrolled subjects regardless of their PD-L1 status and included both squamous and non-squamous disease types. Patients with both squamous and non-squamous disease were randomized (1:1) to receive either atezolizumab administered intravenously at 1200 mg every 3 weeks or docetaxel administered intravenously at 75 mg/m 2 every 3 weeks. The co-primary endpoints were overall survival (OS) in all randomized patients (intention-to-treat [ITT] population) and in a PD-L1-selected subgroup in the primary analysis population. OAK showed that persons assigned to atezolizumab lived a median of 13.8 months, 4.2 months longer than those treated with docetaxel chemotherapy (median overall survival [OS]: 13.8 vs. 9.6 months; HR = 0.74, 95 % confidence interval [CI]: 0.63, 0.87).

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The POPLAR study was a multicenter, open-label, randomized Phase II trial that evaluated the efficacy and safety of atezolizumab compared with chemotherapy (docetaxel) in 287 subjects with previously

treated recurrent locally advanced or metastatic NSCLC. The primary endpoint was OS; secondary endpoints included progression-free survival (PFS), objective response rate (ORR) and safety. From an updated analysis with a median follow-up of 22 months, the median overall survival with atezolizumab was 12.6 months, 2.9 months longer than those treated with docetaxel chemotherapy (median OS: 12.6 vs. 9.7 months; HR = 0.69, 95 % CI: 0.52, 0.92). There were no significant differences between atezolizumab and docetaxel groups with respect to ORR (15% in both groups), complete response (0.7% vs. 0%), and partial response (15% in both groups). Median duration of response in the atezolizumab group was 18.6 months (95% CI: 11.6, not estimable) versus 7.2 months (95 % CI: 5.6, 12.5) in the docetaxel group.

Paraganglioma and Pheochromocytoma

Wang et al (2022) stated that molecular targeted therapy plays an increasingly important role in the treatment of metastatic pheochromocytomas and paragangliomas (PPGLs), which are rare tumors but remain difficult to treat. These investigators provided an overview of established molecular targeted therapies in present use, and perspectives on those currently under development and evaluation in clinical trials. Recently published research articles, guidelines and expert reviews on molecular targeted therapies in PPGLs were systematically examined. Some tyrosine kinase inhibitors (TKIs, e.g., sunitinib and cabozantinib) are already in clinical use with some promising results, but without formal approval for the treatment of PPGLs. Sunitinib is the only therapeutic option that has been examined in a randomized, placebo-controlled clinical trial. It is clinically used as a 1st-, 2nd-, or 3rd-line therapeutic option for the treatment of progressive metastatic PPGLs. Several other promising molecular targeted therapies (e.g., HIF2α inhibitors, tumor vaccination together with check-point inhibitors, anti-angiogenic therapies, and kinase signaling inhibitors) are currently under evaluation in clinical trials. The HIF2α inhibitor belzutifan may prove to be particularly interesting for cluster 1B-/VHL/EPAS1-related PPGLs, while anti-angiogenic therapies appear to be primarily effective in cluster 1A-/SDHx-related PPGLs. Several combination therapies currently being investigated in clinical trials (e.g., temozolomide/olaparib, temozolomide/talazoparib or cabozantinib/atezolizumab) will provide data for novel therapy for metastatic PPGLs. The authors concluded that it is likely that advances in such molecular targeted therapies will play an important role in the future treatment of these tumors, with more personalized therapy options paving the way towards improved therapeutic outcomes.

Prostate Cancer

Vaishampayan (2017) highlighted some developments as well as ongoing challenges of managing advanced prostate cancer. Significant strides are being made in managing metastatic prostate cancer. With the evolution of multiple new therapies, now the optimal use of these therapies and their proper sequencing is being addressed. Research is ongoing for mapping out pathways of resistance to therapies and for discovering new targets. Genomic alterations and abnormalities in circulating tumor DNA are

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being detected and will hopefully lead toward biomarker-based therapies. The next era in oncology belongs to immune therapy. However, in prostate cancer the immune checkpoint inhibitors have shown modest responses and a phase-III clinical trial of radiation therapy ± ipilimumab revealed no benefit. Efforts are ongoing with combination trials of enzalutamide and atezolizumab or pembrolizumab; inhibitors of the enzyme poly ADP ribose polymerase (PARP) are gradually being established for therapeutic purposes, with olaparib achieving breakthrough status for prostate cancer patients with BRCA1 and 2 and ATM mutations. The authors concluded that the future will bring an era of personalized medicine in advanced prostate cancer as well as optimization and more strategic sequencing of existing therapies.

Renal Cell Carcinoma

Atkins and associates (2017) noted that there has been dramatic expansion of the treatment armamentarium for patients with advanced RCC (aRCC), including drugs targeting vascular endothelial growth factor (VEGF) and mammalian target of rapamycin (mTOR) pathways. Despite these advances, patient outcomes remain suboptimal, underscoring the need for therapeutic interventions with novel mechanisms of action. The advent of immunotherapy with checkpoint inhibitors has led to significant changes in the treatment landscape for several solid malignancies. Specifically, drugs targeting the PD-1 and cytotoxic T-lymphocyte associated antigen (CTLA-4) pathways have demonstrated considerable clinical efficacy and gained regulatory approval as single-agent or combination therapy for the treatment of patients with metastatic melanoma, NSCLC, aRCC, advanced head and neck squamous cell carcinoma (HNSCC) , urothelial cancer and Hodgkin lymphoma. In aRCC, the PD-1 inhibitor nivolumab was approved in both the U.S. and Europe for the treatment of patients who have received prior therapy, based on improved OS compared with the mTOR inhibitor everolimus. Other checkpoint inhibitors, including the CTLA-4 inhibitor ipilimumab in combination with several agents, and the PD-L1 inhibitor atezolizumab, are in various stages of clinical development in patients with aRCC.

Small Cell Lung Cancer

Seeber and colleagues (2017) stated that systemic therapy options for small cell lung cancer (SCLC) patients with extensive disease remain poor. After an initial response on 1st-line therapy, virtually all SCLC patients develop disease progression. For those who showed an initial response only few therapeutic options with low response rates are currently available. Until now, many experimental and targeted agents have failed to yield convincing clinical benefits, and new therapeutic options are clearly needed for these patients. The authors noted that in this year's oncological congresses, several new therapy strategies, including checkpoint inhibition, showed promising results in ongoing trials. Furthermore, a potential benefit of new agents targeting DLL3, Aurora A kinase and PARP-inhibitor was reported. These researchers summarized new developments and highlighted the most important and promising data in the treatment of patients with relapsed SCLC; atezolizumab was one of the keywords of this article.

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Horn et al (2018) stated that enhancing tumor-specific T-cell immunity by inhibiting programmed death ligand 1 (PD-L1)-programmed death 1 (PD-1) signaling has shown promise in the treatment of

extensive-stage small-cell lung cancer. Combining checkpoint inhibition with cytotoxic chemotherapy may have a synergistic effect and improve efficacy. These researchers conducted a double-blind, placebo-controlled, phase-III clinical trial to evaluate atezolizumab plus carboplatin and etoposide in patients with extensive-stage small-cell lung cancer who had not previously received treatment. Patients were randomly assigned in a 1:1 ratio to receive carboplatin and etoposide with either atezolizumab or placebo for four 21-day cycles (induction phase), followed by a maintenance phase during which they received either atezolizumab or placebo (according to the previous random assignment) until they had unacceptable toxic effects, disease progression according to Response Evaluation Criteria in Solid Tumors (RECIST), version 1.1, or no additional clinical benefit. The 2 primary endpoints were investigator-assessed progression-free survival (PFS) and overall survival (OS) in the intention-to-treat (ITT) population. A total of 201 patients were randomly assigned to the atezolizumab group, and 202 patients to the placebo group. At a median follow-up of 13.9 months, the median OS was 12.3 months in the atezolizumab group and 10.3 months in the placebo group (hazard ratio [HR] for death, 0.70; 95 % confidence interval [CI]: 0.54 to 0.91; p = 0.007). The median PFS was 5.2 months and 4.3 months, respectively (hazard ratio [HR] for disease progression or death, 0.77; 95 % CI: 0.62 to 0.96; p = 0.02). The safety profile of atezolizumab plus carboplatin and etoposide was consistent with the previously reported safety profile of the individual agents, with no new findings observed. The authors concluded that the addition of atezolizumab to chemotherapy in the 1st-line treatment of extensive-stage small-cell lung cancer resulted in significantly longer OS and PFS than chemotherapy alone.

In March 2019, the FDA approved atezolizumab (Tecentriq) in combination with carboplatin and etoposide (chemotherapy), for the initial (first-line) treatment of adults with extensive-stage small cell lung cancer (ES-SCLC). This approval was based on results from the phase III, multicenter, double-blind, randomized placebo-controlled IMpower133 study, which showed that Tecentriq in combination with chemotherapy helped people live significantly longer compared to chemotherapy alone (median overall survival [OS]=12.3 vs. 10.3 months; hazard ratio [HR]=0.70, 95% CI: 0.54–0.91; p=0.0069) in the intention-to-treat (ITT) population.The Tecentriq-based combination also significantly reduced the risk of disease worsening or death (progression-free survival, PFS) compared to chemotherapy alone (PFS=5.2 versus 4.3 months; HR=0.77, 95% CI: 0.62-0.96; p=0.017). Safety for the Tecentriq and chemotherapy combination appeared consistent with the known safety profile of Tecentriq (Roche, 2019).

Urothelial Carcinoma - Bladder Cancer

Rosenberg et al (2016) conducted a multicenter, single-arm, two-cohort, phase 2 trial in 486 patients aged 18 or older presenting with inoperable locally advanced or metastatic urothelial carcinoma. Inclusion criteria included a requirement that disease be inoperable locally advanced or metastatic urothelial carcinoma whose disease had progressed after previous platinum-based chemotherapy, adequate hematological and end-organ function, no autoimmune disease or active infections, Eastern

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Cooperative Oncology Group performance status of 0 or 1, and measurable disease defined by response Evaluation Criteria in Solid Tumors version 1.1 (RECIST). Results of this study indicated that treatment

with atezolizumab resulted in a significantly improved RECIST objective response rate for each prespecified immune cell group (p = 0.0001 - 0.0004) as well as in all patient combined analysis (p = 0.0058). The Cancer Genoma Atlas (TCGA) subtypes and mutation load were found to be independently predictive for response to atezolizumab. The investigators concluded that atezolizumab was found to have durable activity and good tolerability in the study population, with increased levels of PD-L1 expression on immune cells. The authors stated that, although this is a Phase II clinical trial, this study is well designed with robust statistical power and highly significant findings in support of the use of atezolizumab for this indication.

The U.S. Food and Drug Administration (FDA) granted an accelerated approval to atezolizumab to be used as a first-line treatment for cisplatin-ineligible patients with locally advanced or metastatic urothelial carcinoma (Genentech, 2017).

The treatment-naive IMvigor210 cohort enrolled 123 patients with cisplatin-ineligible locally advanced or metastatic urothelial carcinoma at 47 locations in North America and Europe between June 9, 2014, and March 30, 2015 (Genentech, 2017). Patients received 1200 mg of atezolizumab IV every three weeks until progression. The primary endpoint was ORR, with secondary outcomes measures including progression-free survival (PFS) and overall survival (OS). Among the 123 patients, 119 received at least one dose of atezolizumab. There was an association between tumor mutation load and response. The median progression-free survival and overall survival were 2.7 months and 15.9 months, respectively. The median duration of response had not yet reached at the time of the FDA approval for this indication. Among patients with PD-L1 expression 5 percent or more (32 patients), the ORR was 28.1 percent, including a CR rate of 6.3 percent. In patients with PD-L1 expression less than 5 percent, the corresponding rates were 21.8 percent and 6.9 percent, respectively.

Although Tecentriq was granted accelerated approval in 2016 for the treatment of prior-platinum treated metastatic urothelial carcinoma (mUC) based on the results from the IMvigor210 study (Cohort 2), continued approval was contingent upon the results of IMvigor211, the original PMR for the prior-platinum treated mUC indication. This study did not meet its primary endpoint of overall survival in the PD-L1 high patient population. Subsequently, in consultation with the U.S. FDA, Roche voluntarily withdrew this indication in March 2021 in recognition of the principles of the Accelerated Approval Program. This decision did not affect other approved indications for Tecentriq (Roche, 2021).

On November 28, 2022, Genentech announced the voluntary withdrawal of U.S. indication of Tecentriq (atezolizumab) for the treatment of adults with locally advanced or metastatic urothelial carcinoma (mUC, bladder cancer) who are not eligible for cisplatin-containing chemotherapy and whose tumors express PD-L1 (PD-L1-stained tumor-infiltrating immune cells covering ≥ 5% of the tumor area) or are not eligible for any platinum-containing chemotherapy regardless of PD-L1 status. This decision was based on a consultation with the U.S. Food and Drug Administration (FDA), in accordance with the requirements of the FDA's Accelerated Approval Program. The Phase III IMvigor130 trial evaluated

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requirements of the FDA's Accelerated Approval Program. The Phase III IMvigor130 trial evaluated Tecentriq plus platinum-based chemotherapy for the first-line treatment of patients with previously untreated advanced bladder cancer. This confirmatory trial was the designated postmarketing requirement (PMR) to convert the accelerated approval to regular approval, however, the trial results failed to meet an overall survival endpoint. Genentech has indicated that it will work with the FDA to complete the withdrawal process and to notify healthcare providers in the US (Genentech, 2022a).

Uveal Melanoma

Algazi and associates (2016) noted that the safety and effectiveness of PD-1 blockade in patients with uveal melanoma has not been well characterized. In a multi-center, comparative study, 58 patients with stage IV uveal melanoma received PD-1 or PD-L1 antibodies between 2009 and 2015 at 9 academic centers. Patients who were evaluable for response were eligible for the analysis. Imaging was performed every 12 weeks and at the investigators' discretion. Safety and clinical efficacy outcomes, including the best overall response, PFS, and OS, were retrospectively determined. Of 56 eligible patients, 48 (86 %) had received prior therapy, and 35 (63 %) had received treatment with ipilimumab; 3 patients had an objective response to ipilimumab, and 8 had stable disease (SD) as their best response; 38 patients (68 %) received pembrolizumab, 16 (29 %) received nivolumab, and 2 (4 %) received atezolizumab. Objective tumor responses were observed in 2 patients for an overall response rate of 3.6 % (95 % CI: 1.8 % to 22.5 %); SD (greater than or equal to 6 months) was observed in 5 patients (9 %). The median PFS was 2.6 months (95 % CI: 2.4 to 2.8 months), and the median OS was 7.6 months (95 % CI: 0.7 to 14.6 months). There was no association between prior treatment with ipilimumab or liver-directed therapy and PFS or OS. Treatment was well-tolerated, and only 1 patient discontinued treatment because of toxicity. The authors concluded that PD-1 and PD-L1 antibodies rarely conferred durable remissions in patients with metastatic uveal melanoma.

Various Experimental Indications

There is insufficient evidence in peer-reviewed published literature to support the use of atezolizumab for the treatment of various malignancies including acute myeloid leukemia, appendiceal adenocarcinoma, asymptomatic myeloma, bone plasmacytoma, cholangiocarcinoma, chronic lymphocytic leukemia, chronic myelomonocytic leukemia, colorectal cancer, fallopian tube carcinoma, gastric cancer, gastro-esophageal junction cancer, glioblastoma, Hodgkin lymphoma, mesothelioma, multiple myeloma, myelodysplastic syndrome, non-Hodgkin's lymphoma, ovarian carcinoma, pancreatic adenocarcinoma, peritoneal carcinoma, polycythemia vera, primary myelofibrosis, prostate cancer, renal cell carcinoma, myxoid/round cell liposarcoma and synovial sarcoma, small lymphocytic lymphoma, smoldering multiple myeloma, thyroid carcinoma, uterine cancer, and uveal melanoma.

Appendix

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Appendix

Per Prescribing Information (Genentech, 2021b), Tecentriq (atezolizumab) should either be withheld or permanently discontinued for the diagnosis listed in the tables below:

Table: Diagnoses for which atezolizumab should be withheld

Diagnoses
Pneumonitis Grade 2*
Colitis Grades 2 or 3*
Hepatitis with no tumor involvement of the liver and AST or ALT increases to more than 3 and up to 8 times ULN or total bilirubin increases to more than 1.5 and up to 3 times ULN.*
Hepatitis with tumor involvement of the liver** and baseline AST or ALT is more than 1 and up to 3 times ULN and increases to more than 5 and up to 10 times ULN or baseline AST or ALT is more than 3 and up to 5 times ULN and increases to more than 8 and up to 10 times ULN.*
Endocrinopathies Grades 3 or 4 (e.g., hyophysitis, adrenal insufficiency, hyperthyroidism, and type 1 diabetes mellitus) withhold until clinically stable or permanently discontinue depending on severity.
Nephritis with Renal Dysfunction Grades 2 or 3 increased blood creatinine.*
Exfoliative Dermatologic Conditions: suspected SJS, TEN, or DRESS
Neurological Toxicities Grade 2*
Infusion-Related Reactions Grades 1 or 2 interrupt or slow the rate of infusion

* Resume in persons with complete or partial resolution (Grade 0 to 1) after corticosteroid taper. Permanently discontinue if no complete or partial resolution within 12 weeks of initiating steroids or inability to reduce prednisone to 10 mg per day or less (or equivalent) within 12 weeks of initiating steroids.

** If AST and ALT are less than or equal to ULN at baseline, withhold or permanently discontinue Tecentriq based on recommendations for hepatitis with no liver involvement

ALT = alanine aminotransferase, AST = aspartate aminotransferase, ULN = upper limit normal, DRESS = Drug Rash with Eosinophilia and Systemic Symptoms, SJS = Stevens Johnson syndrome, TEN = toxic epidermal necrolysis

Table: Diagnosis for which atezolizumab should be permanently discontinued

Diagnoses
Pneumonitis Grade 3 or 4
Colitis Grade 4

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Colitis Grade 4

Hepatitis with no tumor involvement of the liver and AST or ALT increases to more than 8 times ULN or total bilirubin increases to more than 3 times ULN.
Hepatitis with tumor involvement of the liver*** and AST or ALT increases to more than 10 times ULN or total bilirubin increases to more than 3 times ULN.
Nephritis with Renal Dysfunction Grade 4 increased blood creatinine.
Exfoliative Dermatologic Conditions confirmed SJS, TEN, or DRESS.
Myocarditis Grades 2, 3, or 4
Neurological Toxicities Grades 3 or 4
Infusion-related Reactions Grade 3 or 4
Permanently discontinue if no complete or partial resolution within 12 weeks of initiating steroids or inability to reduce prednisone to 10 mg per day or less (or equivalent) within 12 weeks of initiating steroids.

*** If AST and ALT are less than or equal to ULN at baseline, withhold or permanently discontinue Tecentriq based on recommendations for hepatitis with no liver involvement.

ALT = alanine aminotransferase, AST = aspartate aminotransferase, ULN = upper limit normal, DRESS = Drug Rash with Eosinophilia and Systemic Symptoms, SJS = Stevens Johnson syndrome, TEN = toxic epidermal necrolysis

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